Ibri Sewer, TE & STP — Project 2621

W4 — Sewer Network Design Pipeline

Methodology and Test-Boundary Results

Internal working document — design team

18 August 2026

Item	Value
Test area	551 ha
Network designed	1,655 chambers / 89.5 km
Peak flow at outfall	83 L/s (Qadf 3,070 m3/d)
Structural rules	loop-free PASS · one outlet per chamber PASS
Audit	2 residual violations
Code / tests	W4/py/sewnet — 56 pytest cases, Table-11 gate

Contents

Executive summary

We built the sewer design pipeline, proved it end to end on the 551 ha test boundary, put it through a 21-agent adversarial review and a two-rule structural audit (no loops; one outlet per chamber), fixed everything both found, and the design now holds those rules with 2 failing checks of 21 out of 1,655 chambers. Here is the whole story in one page.

The pipeline takes four inputs — roads, classified plots, the 0.5 m terrain and a boundary — and produces a complete gravity network: 1,655 chambers, 89.5 km of pipe (95% DN200, stepping up to DN600 on the outfall leg), designed inverts everywhere, about 9 seconds to run. Every one of the 2,987 loaded units (2,217 built plots, 522 planned, 248 unparceled buildings; 86 farms excluded per doctrine) lands on exactly one chamber — nothing silently dropped, mass balance closes exactly. Saturation flow at the outfall: Qadf 3,070 m3/d, peak 83 L/s (Merrimack).

The hydraulics were never taken on faith. The Colebrook-White solver had to reproduce all nine minimum gradients of G203 Table 11 (plus/minus 5%) before it was allowed to size anything — it does, mean deviation under 2%. The adversarial panel then attacked the code and confirmed 11 real defects, all fixed: the spanning tree skipped 22 km of cross-street edges (now covered with a crest-manhole layout); capacity was computed on nominal diameter while OD-designated PVC-U bores smaller (now the true SDR34 bore); a broken bisection in the velocity-cap solver; drop structures measured against the wrong datum. 44 pytest cases lock all of it in.

The structural audit found what the graph checks had hidden (section 5): the tree gave every node one outlet, but 174 chambers sat within 3 m of another chamber — many at the same point — each with its own outlet. On the ground that is a two-outlet junction, exactly what the rule forbids. The pipeline now merges coincident chambers, re-derives the tree over the merged set (which also removes loops that merging exposes), and offsets every extra outgoing pipe so it starts at the next house connection or 10 m clear of the chamber — the SWNETWROK convention, which had not been adopted until it was called for.

The honest self-cleansing picture, on the table rather than under it: 98% of pipes cannot reach 0.75 m/s at saturation peak — physically inevitable on small residential branches — and comply through the guideline's own tractive-force alternative at tau = 1 Pa, a pending assumption [GAP-9]. If NWS sets tau = 2 Pa, 1,160 pipes need steeper slopes. That single number is the strongest argument for pinning tau down at the kickoff.

Three findings from the test area:

- **The outfall landed 733 m from where you expected —** the terrain's lowest boundary road node is at (450614, 2567397), south-center on the main corridor, z 351.2 m, not the west edge. If the real trunk connection is elsewhere, it is one config line and an 11-second re-run.
- **Your elevated-roads concern was justified:** 327 units (10.9%) could not gravity-connect at standard sewer depth. Deepening 236 chambers recovered all but 6, now flagged as local-solution candidates.
- **2 SLS pockets appeared** — all small enough to absorb into detail design per rule 9 (some carry no properties at all and are depth artefacts near the wadi bank rather than real station sites). 351 drop structures (224 vortex-class) concentrate at wadi-bank crossings, exactly where detail design applies the G1-p85 rules.

Also delivered: T01 Rev 3 — the tutorial now teaches Colebrook-White (section 14), Table 11 derived step by step, every number verified.

What is proven: doctrine loads in, guideline-compliant network out, auditable and re-runnable, honest about its assumptions. What is not yet proven: SewerGEMS agreement (package and referee table await the ModelBuilder run), and behaviour at 36-zone scale with the finalised trunk. That is W5.

1. What the pipeline is

A re-runnable Python package (sewnet) that designs a gravity sewer network inside any boundary, given roads, loaded plots, terrain and an outfall or connection point. One config file holds the paths; criteria.py holds every design number with its PAM-GUD page reference — no number lives anywhere else in the code.

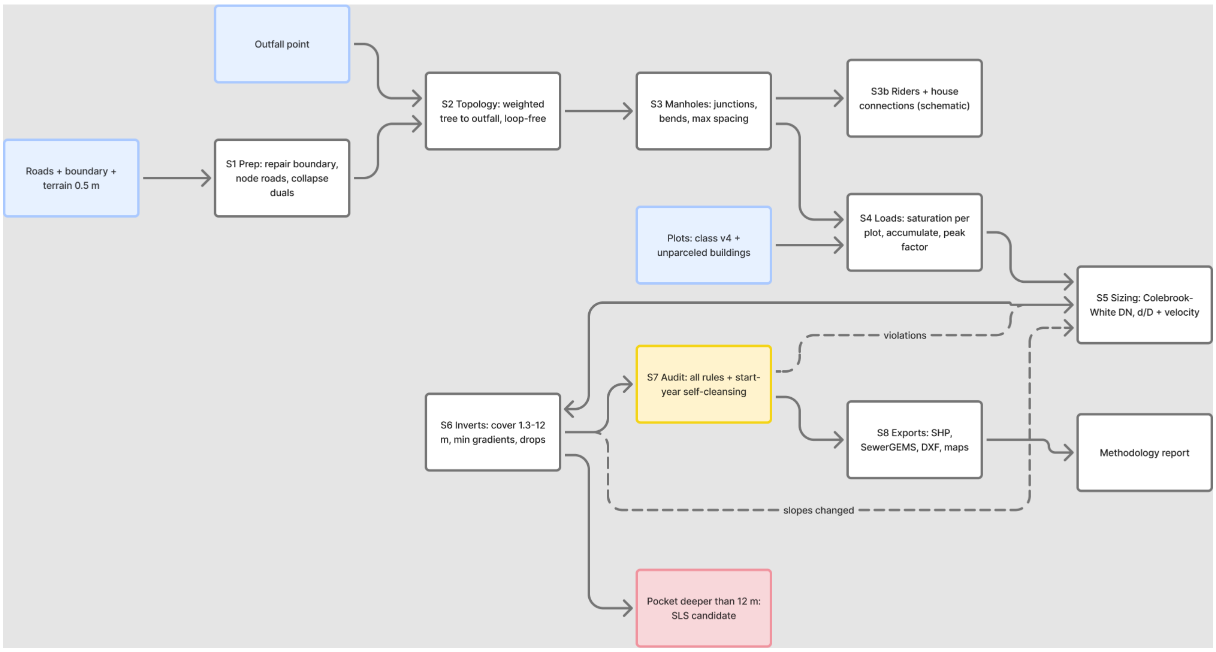

Figure 1 — Pipeline architecture. Blue = inputs, yellow = audit gate, red = SLS flag, dashed = iteration loops.

Stages, in one breath: repair and clip the inputs; node the roads and collapse dual carriageways; grow a loop-free tree toward the outfall (climb-penalised, arterial-preferring — the 'no alleys' lesson); add cross-street branches where loaded plots front an off-tree street; place chambers (junctions, bends over 45 degrees, 100 m spacing); merge coincident chambers and offset extra branch starts; load every plot at saturation; accumulate flows with peak factor and infiltration; size pipes and solve inverts together on the true bore; check every house can reach its chamber, deepening where roads are elevated; audit everything independently; export SHP, SewerGEMS, DXF and maps.

2. From road centrelines to sewer corridors

Raw centrelines are not a sewer corridor. Fed directly, the pipeline put a chamber wherever the survey data happened to break: 576 of 2,137 chambers (27 %) sat at near-collinear breaks, 465 of them at under two degrees of deflection. Those are artefacts of the data, not design decisions, and every one costs a manhole. A separate treatment stage now turns centrelines into corridors before anything is designed.

Treatment	Why	Test area
De-duplicate vertices	zero-length steps corrupt direction measurement — they made 66 reaches read as 180 degree bends in the first compliance check	269 removed
Simplify (0.5 m)	survey jitter, without moving the centreline	applied
Dissolve collinear breaks (under 10 degrees at a two-way node)	a straight street broken into three pieces becomes ONE corridor, so no chamber is placed at the breaks	721 joins
Collapse roundabouts (ring under 150 m, circular)	a roundabout is not a corridor; the ring is removed and its legs reattach to the centre	12 collapsed
Drop dangling stubs under 8 m with no frontage	clip debris that would otherwise become a branch	1 dropped

Treatment	Why	Test area
Classify main roads	main roads cannot be opened longitudinally; crossings remain allowed (G1-p85 trenchless). DERIVED here because the road layer carries no class attribute — advisory until a hierarchy layer is supplied	38 segments flagged

Result: 2165 raw segments become 1298 corridors, and the length barely moves (98.75 to 98.19 km) — the stage re-joins rather than deletes. The corridors are written to W4/shp/W4_corridors.shp with a MAIN_ROAD and ELIGIBLE column so the corridor decisions can be reviewed and hand-edited in QGIS before any design runs.

Effect on the design: chambers fall from 2,137 to 1,655 (about 27 % fewer) with the network length essentially unchanged, so the saving is manholes that were never justified rather than sewers that are now missing.

3. Hydraulic basis and how it is verified

Element	Basis	Verification
Capacity / velocity	Colebrook-White, ks = 1.5 mm, nu = 1.141e-6 m2/s (G203-p24/25/28), partial-full circular geometry, true internal bore (PVC-U OD-series derated to SDR34 ID; GRP nominal = ID)	Table-11 gate: reproduce all 9 minimum gradients at 0.75 m/s within 5% — passes at under 2% mean deviation; ID convention unit-tested
Minimum gradients	Steeper of Table 11 (p29) and tractive force Smin = 2.33e-4 x tau^1.23 x Q^-0.461 (p27, A9-corrected), plus a 40 mm total-fall guard per reach (p29 4.3.1)	tau = 1 Pa tagged [GAP-9]; Q floored at Mara's 1.5 L/s minimum design flow — at the floor tractive is about Table 11 DN200: the methods meet
d/D limits	at or below 0.65 (D<=350) and 0.50 (D>350) at peak (p27 Tab 10)	enforced in sizing, re-checked in audit on the true bore
Velocity band	at least 0.75 m/s at peak or tractive-compliant (p26-27); at most 3.0 m/s via a slope cap whose surplus fall becomes a designed drop	audit plus the transparency statistics in section 6
Loads	Doctrine 2: every plot at saturation, 6.0 x 171.3 l/c/d = about 1.03 m3/d/plot; farms zero; infiltration 720 L/d/km unpeaked; PF Merrimack (mandatory above 100 properties), held at its 100-property value below; Peltier comparison held the same way	mass balance to the outfall closes exactly; unknown CLASS values can never vanish silently

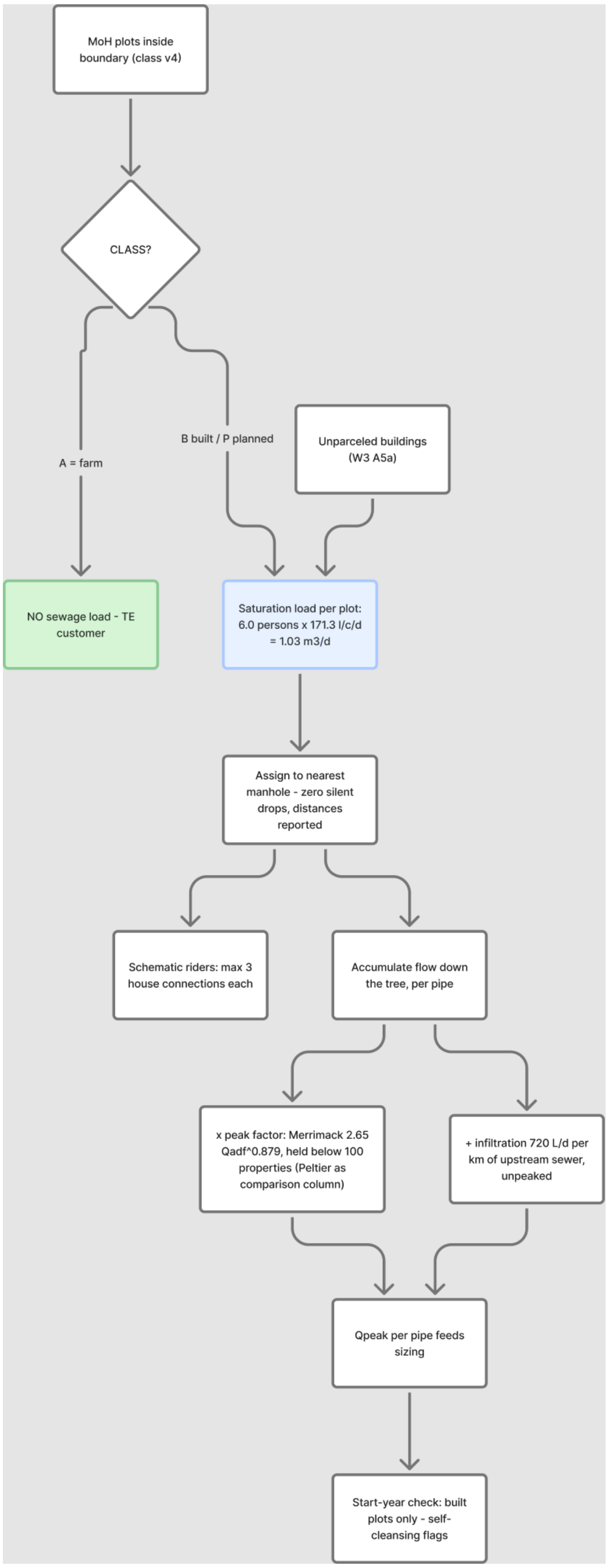

Figure 2 — Load allocation chain.

4. How the solver designs a pipe

Two passes per reach, iterated until no diameter changes (2 iterations sufficed; oscillation between adjacent DNs is detected and broken upward; a final lay pass always leaves inverts consistent with final diameters).

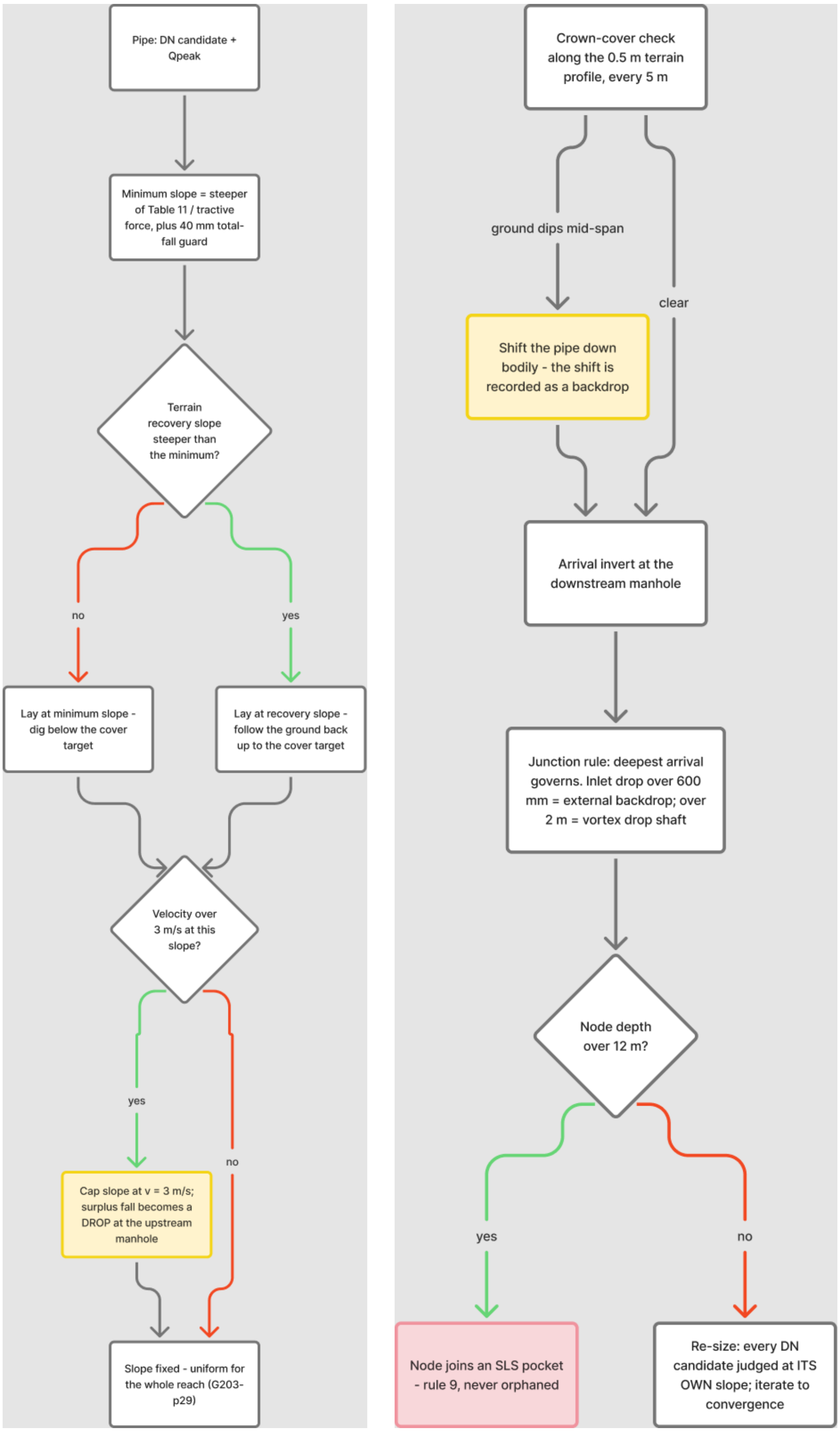

Figure 3 — Solver logic: step 1 sets the reach slope, step 2 resolves profile, junction and depth.

The details that matter:

- **Uniform slope per reach (p29) —** one straight line between chambers, never kinked.
- **The chamber datum is the outgoing invert —** chamber depth is construction depth, and every inlet drop is measured inlet-invert minus outgoing-invert, so velocity-cap surpluses and cover shifts combine into the drop they physically are. Over 600 mm = external backdrop, over 2 m = vortex shaft (p30); the audit re-derives every drop independently, and a missing or misclassified record is a violation.
- **Mid-span cover on real terrain:** every reach profiled at 5 m on the 0.5 m VRT; a reach riding above dipping ground is shifted down bodily, the shift surfacing as a recorded drop.
- **Sizing without the ratchet:** every candidate DN is judged at its own governing slope (no-oversizing, p29), on its true bore, with velocity-infeasible candidates skipped explicitly.
- **Over 12 m depth becomes an SLS pocket (p33 + rule 9) —** never a silent orphan.

5. Loop-free, and one outlet per chamber

Both rules are enforced constructively, not checked after the fact, and both are verified independently on the final output.

No loops. The collection network is a spanning tree by construction: a cost-weighted shortest-path tree to the outfall, where each chamber's single outgoing pipe is its next hop. Loop-closing street edges can never become pipes in that step. Final output: 1,655 chambers, 1,654 pipes, one connected component, acyclic — the pipe count being chambers-minus-one is the arithmetic signature of a tree.

One outlet per chamber. A junction takes any number of inlets and exactly one outgoing pipe. The catch the graph checks missed: two separate chambers can sit at the same physical point (road noding rounds to the centimetre, and the cross-street augmentation planted branch heads next to existing junctions). Each had one outlet in the graph; together they were a two-outlet junction on the ground. The fix, in order:

- **merge every cluster of chambers within 3 m —** they are one structure, which makes the hidden fan-outs visible as chambers with two outgoing pipes;
- **re-derive the tree over the merged pipe set,** because merging can also close a loop (two chambers at one point may additionally be linked by a path). This restores one-outlet-and-no-loops in a single step and marks the leftover pipes as loop-closers;
- **offset every leftover pipe so it starts clear of the chamber:** at the next house connection along its own alignment, or 10 m when that connection sits nearer — the SWNETWROK FANOUT_GAP_M convention. A branch whose pipe is too short to keep a clear start is dropped and reported (its street is served from the far end);
- **slide any remaining branch head clear of neighbouring chambers,** iterating until nothing moves.

Test-area result: 174 coincident chambers merged, 108 fan-outs resolved, 336 branch starts offset, 59 branches dropped as served-from-far-end. Independent verification on the exported network: loop-free PASS, one-outlet PASS, 568 of 569 branch heads at least 10 m clear, one chamber pair at 2.78 m instead of 3.0 m. Those residuals are reported rather than hidden — both are local layout details for detail design.

The same checks now run in the audit every time (mh-clearance, head-offset, one-outlet), with a unit test on a synthetic two-chambers-at-one-point case, so this cannot regress silently.

6. The house-connectability check

Roads are locally elevated for flood protection and underpasses, so houses can sit below the sewer. For every loaded unit: plot ground (0.5 m terrain) minus 0.6 m outlet depth must reach its chamber's invert with 2% fall over the connection distance. Failures raise a deepening requirement on that chamber (capped 0.5 m short of the 12 m limit) and the invert solve re-runs; anything still failing is flagged for a local solution. Test area: 327 flagged, 236 chambers deepened, 6 residual.

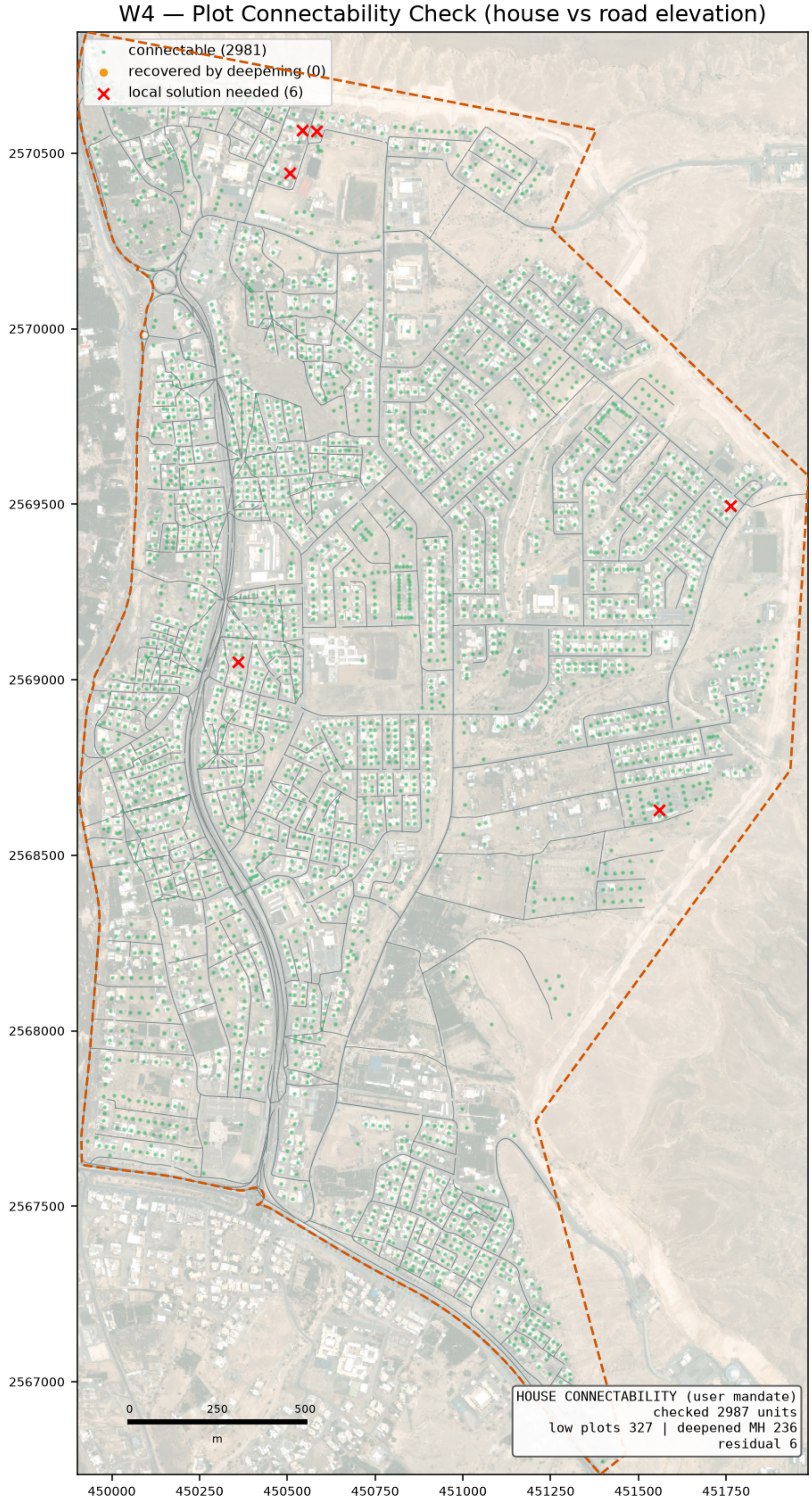

Figure 4 — Plot connectability: green connectable, amber recovered by deepening, red needs a local solution.

7. Test-boundary results

Quantity	Value
Area / roads / network	551 ha / 98.7 km roads / 89.5 km sewers (22.99 km from cross-street augmentation; 0.35 km of unloaded streets deliberately unsewered)
Chambers / pipes	1,655 / 1,654 (single tree, one outfall — chambers minus one)
Diameters	DN200 85.407 km · DN250 1.518 km · DN315 0.981 km · DN400 0.888 km · DN500 0.411 km · DN600 0.331 km
Loaded units	2,987 (built 2,217 + planned 522 + unparceled 248); farms excluded 86
Outfall	(450614, 2567397), z 351.24 — 733 m from user expectation
Qadf / Qpeak at outfall	3,070 m3/d / 83 L/s (PF merrimack; Peltier column in the shapefiles)
Structure rules (section 5)	174 coincident chambers merged · 108 fan-outs resolved · 336 branch starts offset · 59 branches dropped
Depths	max 13.4 m (inside an SLS pocket); gravity network otherwise at or under 12 m
Drops	351 designed structures; 224 vortex-class (over 2 m), concentrated at wadi-bank crossings
SLS pockets	2 (about 5 properties total, absorb per rule 9)
Low plots	327 flagged, 6 residual after deepening 236 chambers
Audit	2 violations — structural residuals only (one chamber pair at 2.78 m vs the 3.0 m clearance; one branch head at 9.1 m vs the 10 m offset); everything else clean, including independent drop re-derivation
Self-cleansing transparency	1,616/1,654 pipes below 0.75 m/s at saturation peak — compliant via tractive at tau=1 Pa [GAP-9]; 1,160 would need redesign at tau=2 Pa; start-year flags 1,640, all tractive-compliant
Runtime	about 11 s design + 4 s exports

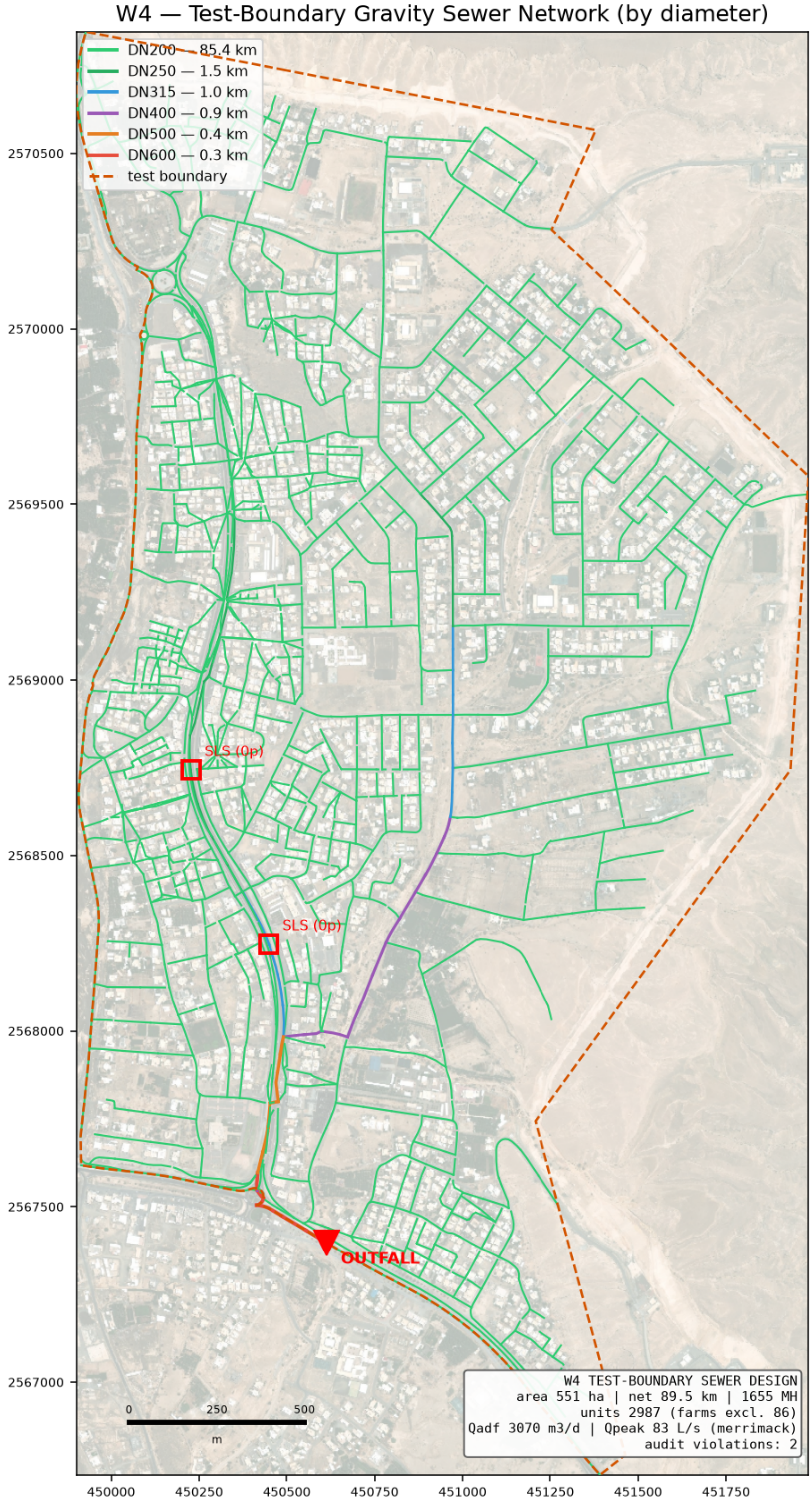

Figure 5 — Designed network by diameter, with outfall and SLS candidates.

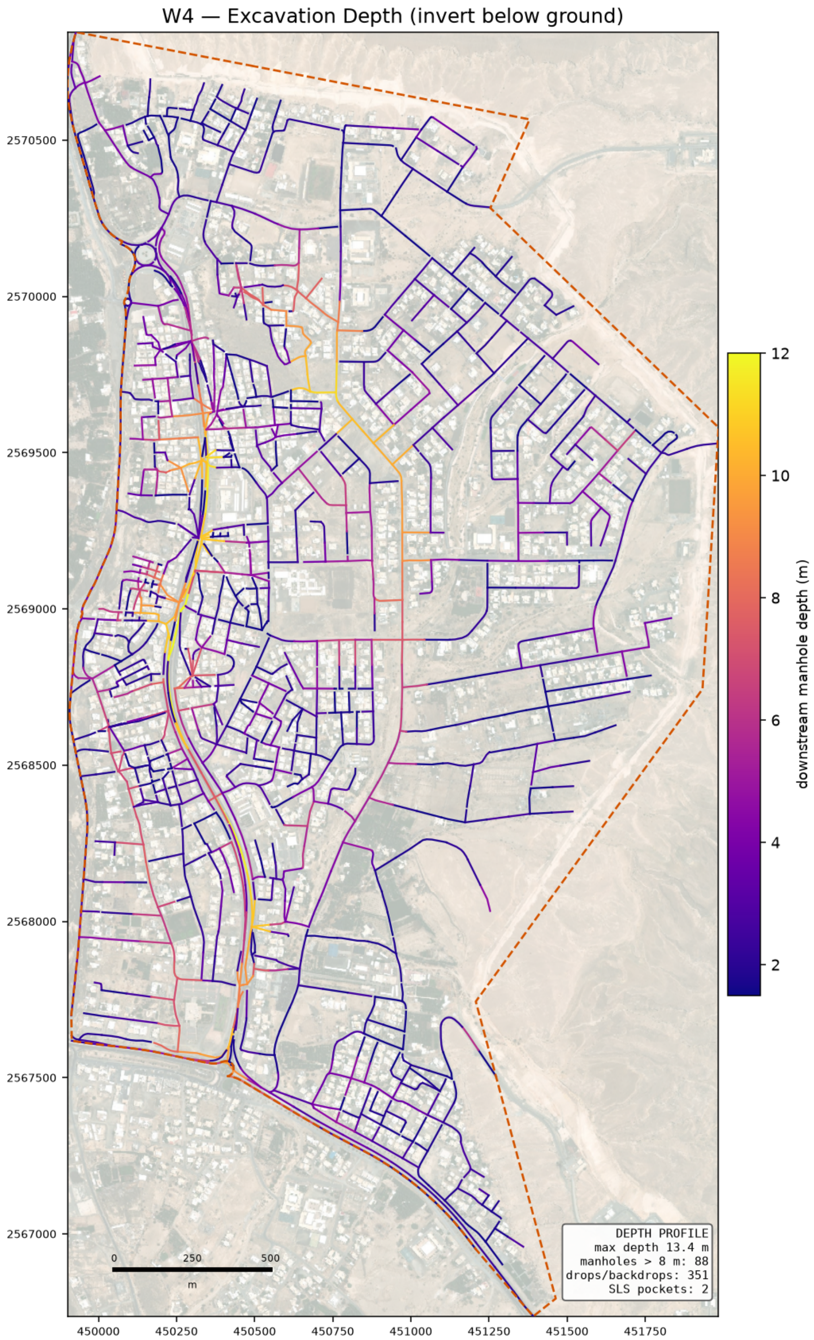

Figure 6 — Excavation depth (invert below ground).

8. SewerGEMS package and referee protocol

W4/sewergems/ holds MANHOLES, CONDUITS and OUTFALL shapefiles built to the Bentley-documented ModelBuilder mappings (explicit START_ND/STOP_ND plus vertex-snapped geometry digitised upstream to downstream, elevations not depths, numeric mm diameters), LOADS.xlsx (pattern-based rows, L/s per chamber), and IMPORT_PROCEDURE.md with the known traps — the 'Set Invert to Start/Stop Node = False' global edit being the one that silently deletes drop manholes. Manning n is exported as 0.013, the ks = 1.5 mm equivalent: the Tab-8 range 0.009-0.011 conflicts with the ks mandate and would show roughly 30% phantom capacity, so that conflict is flagged as an NWS kickoff item.

After the model run, paste SewerGEMS discharge, velocity and d/D into REFEREE_pipes.csv; any pipe deviating more than 5% from our columns is an open investigation. The design is not verified until the two engines agree — deliberately a separate run, not a self-check.

9. Assumptions register

Tagged in criteria.ASSUMPTIONS with the same wording, and reported in every deliverable:

Assumption	Basis / exposure
tau = 1 Pa	GUD-203 gives no numeric design tractive stress [GAP-9]; largest redesign exposure (1,160 pipes at tau=2)
Tractive Q-floor 1.5 L/s	Mara simplified-sewerage minimum design flow; unfloored the formula demands unbounded slopes as Q approaches 0
PVC-U wall class SDR34/SN8	true bore for hydraulics; actual class per PAM-SPC-207 pending
OR 6.0, 1 property per plot	GAP-5 (NCSI housing units missing)
Plot outlet 0.6 m, 2% connection fall	method choice for the connectability check
40 m frontage rule	an off-tree street gets a sewer when a loaded unit lies within 40 m
10 m branch-start offset, 3 m chamber clearance	layout conventions from SWNETWROK and the user's rule, not PAM-GUD values
Chamber size ladder	02 has no size table; GUD-203 4.4 re-extract pending (no hydraulic effect at concept)
Rider discharge to nearest chamber	02 silent on the discharge point
Gravity in-road position	taken from the force-main clause (p51, A9) as an inference
PF held below 100 properties	G1-p71 prescribes no formula there; Peltier held the same way
Infiltration unpeaked	add-order not stated in GUD-201 — kickoff item

10. Limitations and what changes at full scale

- **The trunk is user-finalised (settled 2026-08-18) —** the pipeline designs subnetworks into the given connection points, which are config entries, not structure.
- **One outfall per run today;** multi-connection territory competition is the W5 structural addition.
- **Riders are schematic;** junction losses ignored (normal-depth hydraulics) — both noted for the SewerGEMS comparison.
- **The 0.5 m terrain is concept-grade for inverts near the 0.75-1.0 mm/m minimums (G1-p36);** flat trunk profiles at scale need survey-grade data — a registered data request.
- **tau = 1 Pa carries the largest redesign exposure —** pin it at kickoff.

11. Adversarial review and structural audit

A 21-agent skeptic panel attacked the hydraulic core after the first audit-clean run: four attack lenses (Colebrook-White implementation, solver clause compliance, load and audit doctrine, executed edge cases) followed by independent verification of every raw finding. 17 raw findings, 11 confirmed, 11 fixed, 6 refuted. The headline four were cross-street coverage, the true PVC bore correction, the drop-datum correction, and the velocity-cap bisection repair.

A second, user-driven audit then checked the two layout rules directly against the run output: no loops (held) and one outlet per junction (held in the graph, broken on the ground — see section 5). The 10 m branch-start offset had not been implemented at all; it now is, with permanent audit checks and a unit test. Both audits are recorded in W4/docs/REVIEW_FINDINGS.md.